

CWIP: Coupling With Interpolation Parallel Interface

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Summary

CWIP is an open-source library for the parallel coupling of independent simulation codes in multi-physics research. It provides MPI-based communication and fully distributed data exchange on non-coincident meshes, including on-the-fly spatial interpolation. Unlike other frameworks such as MUI ([Tang et al., 2015](#)) and preCICE ([Chourdakis et al., 2022](#)), CWIP focuses on spatial interpolation and allows users to leverage solver-specific numerical methods, including finite volumes, finite elements, discontinuous Galerkin, and spectral methods. Advanced users can combine CWIP's geometric algorithms with solvers' internal interpolation through callbacks, improving accuracy across heterogeneous solvers.

CWIP supports a wide range of interface geometries, from linear, surface and volumetric elements to higher-order elements with curved edges and faces. Performance-critical algorithms have been optimized for dynamic and adaptive meshes. Recent developments in the ParaDiGM library ([Quémerais et al., 2026](#)) provide dynamic load and memory balancing for large-scale simulations. Ongoing work includes GPU acceleration of geometric algorithms ([Cazalbou et al., 2024](#)) and the development of complementary temporal interpolation methods ([Simon et al., 2025](#)).

Statement of Need

Multi-physics problems can be approached either through monolithic methods — where all physical phenomena are solved simultaneously within a single solver — or through partitioned methods, where independent solvers exchange data across shared interfaces. Partitioned approaches are particularly attractive in high-performance computing (HPC) contexts, as they allow the reuse of existing specialized codes while maintaining their native parallel architecture.

Robust and efficient coupling tools are essential to enable data exchange between solvers operating on distinct meshes, geometries, and discretization schemes. CWIP was initially developed in 2009 to provide an open-source solution complementary to MPCCI ([Joppich & Kürschner, 2006](#)), whose black-box nature limited flexibility and transparency for multi-physics research.

A key strength of CWIP lies in its ability to **adapt to solvers' own numerical methods** — finite volumes, finite elements, discontinuous Galerkin, or spectral methods — and to leverage their internal interpolation capabilities. In an advanced mode, CWIP exposes the results of its geometric algorithms, allowing users to combine them with the solver's internal methods via a *callback*. For example, it is possible to use the functional bases of high-order elements from the source mesh while knowing the coordinates of target points from a finite-difference solver ([Léger et al., 2012](#)), improving accuracy between heterogeneous solvers while maintaining consistency with each solver's numerical scheme.

State of the Field

Existing coupling frameworks differ significantly in scope, geometry support, and interpolation capabilities. MUI (Tang et al., 2015) primarily supports point clouds and focuses on temporal and spatial interpolation in a general-purpose fashion. preCICE (Chourdakis et al., 2022) supports point clouds or meshes composed of planar triangular or quadrilateral elements and provides a broad ecosystem of adapters for established simulation codes. OpenPALM (Duchaine et al., 2015) is a higher-level coupler that relies on CWIPI as its interpolation engine.

CWIPI addresses several limitations of these frameworks. It supports a broader range of interface representations, including linear, surface and volumetric elements, polygons and polyhedra, as well as higher-order elements with curved edges and faces. Where preCICE (Chourdakis et al., 2022) and MUI (Tang et al., 2015) focus on generic coupling workflows, CWIPI focuses primarily on spatial interpolation fidelity, leaving temporal interpolation strategies to the user's implementation — an approach well suited to research contexts where solver-specific accuracy is paramount.

Software Design

Since version 1.0, CWIPI is built on top of the ParaDiGM library (Quémerais et al., 2026), originally created specifically to meet the needs of CWIPI. ParaDiGM provides the low-level geometric algorithms and parallel data structures. The coupling workflow is organized around three main components: geometric localization (finding which source mesh element contains each target point), data redistribution (MPI-based parallel exchange of located data), and interpolation (computing field values at target locations from source element data).

CWIPI provides three APIs — C/C++, Fortran, and Python/NumPy — making it compatible with a wide range of scientific codes. The integration of CWIPI into an existing solver is designed to be **minimally intrusive**: it does not require modifying the code structure, and the API relies solely on simple arrays, keeping the coupling layer lightweight and straightforward to adopt.

A major design challenge lies in the **performance of geometric algorithms**, especially when the coupling interface moves during the simulation or when solvers employ dynamic mesh adaptation. In the 0.x branch of CWIPI, algorithms relied on the FVM library from Code_Saturne (Archambeau et al., 2004), which later evolved into PLE, a lightweight coupling library (Fournier, 2020). New algorithms (Andrieu et al., 2026) developed within ParaDiGM provide **dynamic load and memory balancing** at each coupling step, which is critical because coupling problems are inherently unbalanced across MPI processes. These developments bring significant performance gains for large-scale and complex simulations.

Future Directions

Two main research directions are currently pursued. The first concerns the **GPU porting** of the most computationally expensive geometric algorithms, for which new algorithms have recently been designed (Cazalbou et al., 2024) and will be integrated in an upcoming release. The second focuses on the development of **temporal interpolation methods** (François & Massot, 2023; Simon et al., 2025), complementary to those provided by existing frameworks such as preCICE, aiming to improve accuracy and flexibility for dynamic interfaces and heterogeneous solvers.

Research Impact Statement

From a scientific perspective, CWIPI is primarily used within the aeronautics and aerospace community from which it originated. The library is integrated into several codes developed

84 at ONERA, including elsA (Cambier et al., 2013) for computational fluid dynamics, CEDRE
85 (Refloch et al., 2011) for multiphysics simulations, Z-set (Garaud et al., 2019) for structural
86 and materials mechanics, MoDeTheC (Dellinger et al., 2024) for thermal degradation of
87 materials, and SPACE (Delorme et al., 2005) for acoustic propagation. Through these tools,
88 CWIPI enables advanced multiphysics simulations in aerothermodynamics (Errera et al., 2019),
89 aeroacoustics (Langenais et al., 2019), magnetohydrodynamics (Rocamora et al., 2025), electric
90 arc propagation, and fire safety (Dellinger et al., 2023).

91 CWIPI is also employed as both an internal and external coupling library in leading combustion
92 simulation environments such as YALES2 (Moureau et al., 2011) and AVBP (Schonfeld &
93 Rudgyard, 1999). A notable HPC application (Dombard et al., 2018) combining CWIPI with
94 AVBP on a complex geometry demonstrated the robustness and scalability of the coupling
95 approach on massively parallel architectures. Its capabilities were highlighted as early as 2017
96 (Duchaine et al., 2017) and confirmed by an invitation to the ExCALIBUR workshop (Andrieu
97 & Quémerais, 2021). Recent work in aeroelasticity with YALES2 (Fabbri et al., 2023) further
98 demonstrates its versatility. CWIPI's integration into the software stack of national HPC
99 facilities such as the CCRT confirms its recognition at a national level.

100 Beyond its original ecosystem, the open-source community has adopted CWIPI through GitHub-
101 based interfacing projects with OpenFOAM (Moratilla, 2022; nd-92, 2025; Weller et al.,
102 1998) and Nektar++ (Cantwell et al., 2015), enabling applications in aeroacoustics and data
103 assimilation (Valero, 2026). Finally, CWIPI serves as the interpolation engine of the OpenPALM
104 coupler (Duchaine et al., 2015), illustrating its central role in the broader multiphysics and
105 HPC coupling ecosystem.

106 AI Usage Disclosure

107 Generative AI tools were used to assist with the writing of this paper: translation, formatting,
108 and improving the fluency of the text. No AI was used in the development of the CWIPI
109 software. All scientific content was written and validated by the authors.

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119 Author Contributions

120 The contributions to this software are listed according to the CRediT taxonomy:

- 121 ■ **Eric Quémerais:** Conceptualization, Methodology, Software, Validation, Writing –
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- 123 ■ **Bastien Andrieu:** Software, Validation, Writing – Original Draft
- 124 ■ **Karmijn Hoogveld:** Software, Validation, Writing – Original Draft

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